

The Evolutionary Approach: Change Over Time

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Outline

- Overview
- EP guiding principles
- Intelligence from and EP Perspective

Overview

- How did we become as we are?
- Evolutionary psychology (EP) try to answer this question
- Focus on evolutionary approach to understanding human behavior and the human mind
 - Seek to explain how people made psychological adaptations in order to survive the dangers of the past
- Evolutionary psychology is a scientific discipline that approaches human cognition, emotions, and behavior through a lens that incorporates the effects of evolution over time Combines Biology and Psychology

Overview

- The first perspective, EP, is concerned with how the human mind came into existence.
 - It attempts to describe the forces in our ancestral past that could have given rise to mental structures.
- The second perspective, evolutionary computing (and the related discipline of artificial life), emerged from areas of study in artificial intelligence.
- They look at evolutionary processes as a form of adaptive problem solving's and can help us to understand human mental functioning.
- The third perspective, called Neural Darwinism, has its roots in neuroscience and sees evolutionary forces as being behind the creation of neural circuits.

Evolutionary psychologists argue that many of our cognitive mechanisms are innate.

Overview

What has enabled humans to survive and thrive in a harsh world?

Overview

- Homo sapiens possesses the most sophisticated mental faculties of any species on earth.
- Our capacities to think, to use language, and to solve problems outstrip those of any other animal.
- An advanced mind allows for planning, communication, social cooperation, and the development of technology, which give us a tremendous competitive survival advantage.
- But how did the human mind come into existence?
- What processes formed it?

Overview

- In evolutionary theory, changes in an environment give rise to corresponding physical changes in an animal species
- It is believed that the period of time during which many human psychological mechanisms evolved was the Pleistocene era, which began approximately two million years ago.
- This period is referred to as the Environment of Evolutionary Adaptation (EEA)
- Because we cannot go back in time we must make inferences about the physical and social environment of that time, based on evidence of various types.
 - Archeological records, data from studying hunter-gatherer societies, systematic observation of human behavior, and self-reports in the form of interviews and questionnaires.

Overview

- Those who had more problem-solving skills would be more likely to survive
- Adaptive skills:
 - Process information
 - Solve problems
 - Communication skills
 - Express emotions
 - Identify others' intentions
 - Choose a healthy mate
 - Group cooperation
- Evolutionary psychology suggest that the underlying neural mechanisms are shaped by evolutionary forces.

These adaptations are now **hardwired** into the human genome through the **natural selection** process

EPs believe that these adaptive behaviors have become rooted in deeply complex neural circuits in the brain and that they are inherited by subsequent generations

Evolved Psychological Mechanisms

- Each mental mechanism is assumed to have evolved in response to the need to solve a specific problem of adaptation.
- Buss (1999) defines an evolved psychological mechanism characterized by the following six properties:
 1. An evolved psychological mechanism exists because it contributed to the solution of a specific problem of survival or reproduction
 2. Evolved psychological mechanisms are designed to take in only a narrow band of information
 3. The input of an evolved psychological mechanism communicates to an organism the particular adaptive problem it is facing.
 4. The input of an evolved psychological mechanism is transformed by decision rules into output.
 5. The output of an evolved psychological mechanism can be physiological activity, information that will serve as input for other psychological mechanisms, or a behavior
 6. The output of an evolved psychological mechanism is directed toward the solution to a specific adaptive problem.

EP views the mind not as a “one size fits all” machine, but as a collection of functionally distinct mechanisms.

EP Guiding Principles

- Guiding principles of EP
 - Principle 1: the brain is a physical system.
 - It functions as a computer. Its circuits are designed to generate behavior that is appropriate to your environment circumstances.
 - Principle 2. Our neural circuits were designed by natural selection to solve problems that our ancestors faced during our species' evolutionary history
 - Principle 3. Consciousness is just the tip of the iceberg; most of what goes on in your mind is hidden from you. As a result, your conscious experience can mislead you into thinking that our circuits is simpler than it really is.

EP Guiding Principles

- Guiding principles of EP
 - Principle 4. Different neural circuits are specialized for solving different adaptive problems
 - The circuitry involved in vision (seeing predators) is not the same as for hearing
 - Principle 5. Our modern skulls house a stone age mind
 - Your mind is based on adaptive changes that originated in the Pleistocene era (100,000 to 1.8 million years ago)

EP Guiding Principles- AI Perspective

- EP Principle 1: The brain functions like a computer
 - **Connection to AI:** This is the foundational logic of AI....Just as the brain processes inputs (sensory data) to produce outputs (survival behaviors), AI uses neural networks to process data into predictions or actions.
 - **Example:** A Self-Driving Car. Like a biological brain, it uses "physical" sensors (LiDAR/Cameras) to process environmental input and execute a "survival" behavior: braking to avoid a collision.
- EP Principle: Our neural circuits were designed by natural selection
 - **Connection to AI:** AI is mirrored by Genetic Algorithms or Reinforcement Learning....instead of nature, the "selector" is the programmer or the reward function that "evolves" the model over millions of iterations.
 - **Example:** AlphaGo. It wasn't just given a list of rules; it "evolved" its strategy by playing millions of games against itself.

EP Guiding Principles- AI Perspective

- **EP Principle:** Most of what happens in your mind is hidden.
 - Connection to AI?
 - Example?
- **EP Principle4:** Different neural circuits are specialized for solving different problems
 - Connection to AI?
 - Example?

Evolutionary Theory

- Theory of evolution

Evolution is the process of change in living species that occurs over many generations

What causes the change?

Evolutionary Theory

- Theory of evolution
 - What causes the change?
- Mutations in the genes of individuals of a species, mutations that occur spontaneously.
 - Some mutations have no apparent effect on the organism in which they occur and will never be felt or noticed.
 - Some mutations have a beneficial effect, giving the individual in whom they occur an edge in surviving.

Evolutionary Theory

- A mutation that equips an organism in a particular way to manage in its environment more successfully.
- Natural selection
- Darwin noted the variation in characteristics among species and even within species.
 - This variation serves them well in their different environment

Human Evolution

- To explain the evolution of human species, we depend largely on fossils.
- Carbon and other dating methods allow us to roughly estimate the age of these fossils
- From the type of changes that occurred, we can infer- and sometimes provide evidence..... changes due to mutations, permitted our species to survive despite extreme environmental changes over time.
- Much can be inferred about the development of our sort of brain, human exhibits many mental capacities that are extremely complex and behaviors that seem inexplicable in our current environments.
- Fear of snake: has not been learned by actual confrontation

Human Evolution

- Just as **archaeologists** must infer uses for their ancient findings, **evolutionary psychologists** must make inferences, based on various types of evidence, about the environment
- These inferences result in hypotheses that in the ordinary course of science are then tested.
- How do you test the hypothesis?

“ Once clearly formulated hypothesis about evolved psychological mechanism and associated predictions are specified, the next step is to test them **empirically**?”

Evolution of the Human Mind

- The evolution of the human mind responded problems arising in the environment that required solving by human mental capacities.
- No amount of conceptual work can definitively yield a complete list of all the adaptive problems human faced.
- Identifying the full set of human adaptive problems is an enormous task that will occupy scientists for decades to come
- Number of methods employed
 - Cross-cultural traits that are considered universal can be sought in members of different cultures existing in many different environment

Evolution of the Human Mind

- A universal response of mothers to a threat to their child:
 - Supports the hypothesis that this reaction in the distant past aided survival of the species and therefor spread
- Another way of testing.... use methods developed by neuroscience
 - A possible adaption that allows a human male to recognize, in a child's facial resemblance to himself, the probability that the child is in fact his.
 - We may hypothesis from this a particular difference in human male and female brains. Some support for this hypothesis has been found through the use of fMRI (functional magnetic resonance imaging)
 - fMRI used to demonstrate a possible neural mechanism that may account for the observed sex difference

Intelligence from and EP Perspective

- Intelligence has been approached by some as a capacity that can be **measured in some definable terms** for use in meeting educational needs.
 - Further segmented as certain categories of abilities
- Review of categories
 - **Triarchic theory of intelligence:- goal directed and adaptive** (Sternberg's)
 - Everyday life requires dealing with ... managing conditions that arise in our environments.
 - **Since we constantly face these changing conditions, we must adapt to them or in some way fail to keep up with life's requirements.**
 - This task is carried out in part with the aid of practical intelligence that rests **on tacit knowledge** knowledge that is **not the result of specific teaching**

Intelligence from an EP Perspective

- Using Diffusion Tensor Imaging (DTI) has led to a hypothesis regarding how the two hemispheres of the brain communicate.
- **Male Brain Connectivity:** Hypothesized to have stronger connectivity within each hemisphere (front-to-back).
 - Facilitate better coordination between perception and coordinated action.
- **Female Brain Connectivity:** Hypothesized to have stronger connectivity between the left and right hemispheres (side-to-side).
 - Facilitate better communication between analytical and intuitive processing modes.

Intelligence from and EP Perspective

- Review of categories
 - Gardner's approach.... he identifies and refers to different aspects of our mental abilities as “intelligences”, indicating that they are indeed separate if not independent capacities.
 - Linguistic intelligence, musical intelligence, logical-mathematical intelligence, spatial intelligence, bodily-kinesthetic intelligence, natural intelligence, and intra- and interpersonal intelligences.
 - Leaning heavily on localization of function
 - A stroke in the left frontal lobe can destroy Linguistic intelligence, leaving a person unable to speak. However, that same person might still be able to compose music (Musical intelligence) or solve complex puzzles (Spatial intelligence).
 - Evidence from neuroscience supports this approach
- The field of evolutionary psychology seeks to learn how human intelligence evolved

Intelligence from an EP Perspective

- What is directly observable in our quest for understanding how it all happened is the organ in which we consider our intelligence to be housed.
- **Principles:** the brain is a physical system, functioning as a computer whose circuits are designed to generate behavior that is appropriate to environmental circumstances.
- This principle can be challenged on two points:
 - **Brain functions as a computer** Is oversimplification ... the human brain is not simply a physiological computer
 - **The word designed**.... misleading However design in this context is used to describe the result of natural selection, not intentional planning

Intelligence from an EP Perspective

- Brain size and weight:
 - Remarkably large, both in comparison to those of our most closely related primate relatives and relative to our body size, having tripled in size over the course of its evolution
 - At the same time it constitutes a remarkably small percentage of the average body weight of humans – perhaps 2% to 3% -- given the amount and quality of sustenance it requires.

Intelligence from and EP Perspective

- According to (Leonard & Robertson, 1994)
 - Among primates, the relative proportion of resting metabolic energy used for brain metabolism is positively correlated with relative diet quality.
 - Humans represent the positive extreme, having both a very high quality diet and a large brain that accounts for 20-25% of resting metabolism.
 - Evidence from the hominid fossil record implies that major changes in diet and relative brain metabolism occurred with the emergence of the genus Homo
- The obvious complexity of this unusual brain, a number of hypotheses have been proposed regarding the question of how, or why, it alone took the developmental turn, somewhere during prehistoric times, that led to its present state.

Hominid: a taxonomic family of primates that includes eight extant species in four genera: Pongo; Gorilla; Pan; and Homo

Intelligence from an EP Perspective

- Major hypotheses... ecological dominance/social competition (EDSC)
 - Once humans became so good at surviving the environment (**ecological dominance**), the primary struggle for survival shifted from fighting nature to outsmarting each other (**social competition**).
 - **Formulated a couple of decades ago....** to provide an integrative explanation for the unique traits of humanity.
 - **We are here that our ancestors became increasingly able to cope with these forces of nature, presumably via natural selection.**

Intelligence from an EP Perspective

- Major hypotheses... ecological dominance/social competition (EDSC)
 - The competitive relations today among different groups of people indicate the sorts of social difficulties that may have existed among various groups of our ancestors as they competed for the necessities of life.
 - At least some traits that characterize humans must have been those that aided survival within the social context... resulting in the development of our current brain as a social tool.
 - We respond to kin our immediate family, our ethnic group, or religious sect, even our fraternity brothers and sorority sisters.... different from the way respond to others, to outsiders.

Intelligence from and EP Perspective

- The new forms of intelligence hypothesized by EDSC include:
 - Consciousness
 - Language
 - Self-awareness
 - **Theory of mind**.... The ability to intuit what may be transpiring in someone else's mind.... Facial expression, body language, sensing others' emotions and thoughts
- How does one explain art, music, opera, literature, humor, politics, science or religion, using arguments from biological evolution?

Intelligence from an EP Perspective

- Difficulties rest on principle 5 our modern skulls house a Stone Age mind
- There were indeed attempts at empirical testing of predictions that derive from the EDSC hypothesis.
 - Level of intelligence should increase as the population increases
 - Evidence suggested that 175 hominid crania dating from 1.9 million to 10 thousand years ago, does indicate that larger brain volume correlated with conditions such as climatic challenges and ecological challenges
 - How do you learn whether there is also such a correlation between a larger population and larger brain volume?

Intelligence from an EP Perspective

- Other surprising observations:

- In more recent times the human brain appears to be shrinking

- In sept 2010 issue of Discover Magazine Anthropologist John Hawks is stated that “Over the past 20, 000 years the average volume of the male human brain has decreased from 1,500 to 1,300 cubic centimeter with the female brain volume decreasing in about the same proportion?”

- Why?

Intelligence from and EP Perspective

- Why?
- Explanation proposed by researchers....
 - When population became dense within a given areas, beginning around 15,000 to 10,000 years ago, people did not have to be as smart as before in order to survive.
 - The increase in human population provides a larger gene pool, with a correspondingly greater chance of DNA mutations. ... alter characteristics of future generations.

Localization of Function from an EP Perspective: Modularity

- Researchers in the 18th and 19th centuries approached the issue of the brain's different functions.
- Franz Joseph Gall's three basic principles
 - The brain is the sole organ of the mind
 - Basic character and intellect are innately determined
 - Where there is variation in function there must be variation in the in the structures that control the functions.
- The idea of localization of function led scientific thinking a stray for a while....but the basic notion was valid.

Localization of Function from an EP Perspective: Modularity

- In most recent works, it was noted that the interconnectedness of brain function, and the parallel processing that takes place as many different elements to the brain act simultaneously.
- Brain plasticity observed that areas of the brain normally devoted to other purpose can sometimes take over, at least in part, functions of damaged areas

Localization of Function from an EP Perspective: Modularity

- EP addresses the localization of functions issue
 - **Fourth principles** ... different neural circuits ... domain specific systems--- are specialized for solving different adaptive problems
 - **Specific areas of the brain direct specific bodily acts such as moving a hand or a foot**
- EP applying this principle to the mind as well, hypothesizing “**modules**” ... mental organs that perform specific functions
- The behaviors occasioned by these modules are automatic we are not aware of the processes that underlie our behavior , we just act or respond.
- Case... **prosopagnosia**.... the loss of the ability to recognize familiar faces... this suggested that a face recognition module that can be damaged independently of any other abilities.

Localization of Function from an EP Perspective: Modularity

- Another line of thought, within EP:
 - Perception is for moving around in space, not simply for the sake of knowledge, which allow us to judge distance.
 - The phenomenon of prosopagnosia is called up on with regard to perception ... “ perceptual abilities give some of the clearest examples of the modularity of psychological processes” ... its absence can be taken as evidence for modularity of the brain.
 - The term modularity has been used in the attempt of EP to explain the organization of brain function
 - Modularity controversial and still under construction

Categorization from an EP Perspective

- Categorization necessary to our survival
- Species of all sorts categorize items in their environment in order, for example, to avoid eating what is toxic to them... or for that matter to avoid being eaten themselves
- How elements in our environment are grouped together matters.
- Do we sort by shape? By color? By function? So many choices

Memory from an EP Perspective

- Memory crucial to our survival
- EP perspective... how our memory functions evolved?
 - Memory is very complex
 - When an areas of someone's brain is damaged and a portion of memory is lost, memory for other things is not necessarily affected.
 - Evolutionary sense: if the whole of memory were erased by trauma to a portion of the brain, an individual so afflicted could not possibly survive.

Memory from an EP Perspective

- **Hypotheses....** human memory also evolved to retain information that is important for the physical well-being of the individual.
- **Evolutionary psychologists cite a variety of studies that relate to memory for what counts with regard to survival.**
 - A theory has been proposed that women are better at one sort of spatial memory and men at a different one
 - A number of studies testing the hypothesis that men and women have different degrees of competence with respect to spatial relations.
 - **Researchers prediction:-** in all seven ethnic groups and all 40 countries men scored significantly higher than women on the three-dimensional mental rotation test.

Questions

- What are the EP Principles?
- Discuss intelligence, modularity and memory from the EP perspective?

Philosophical Issues in Cognitive Science

- We do not know the origins of our human mind, but still we seek to understand it
- Research conducted back some 2,500 years.
- Philosophy has a far more extensive history than the other disciplines involved in what is now called cognitive science
- Philosophy has long played a very important role in our search for understanding the universe that confronts us and indeed for understanding ourselves.
- The root of Western Philosophy date back to the early Greeks.
- Philosophy.... Greek language Love of wisdom

Philosophical Issues in Cognitive Science

- Once thought of as the queen of the science, philosophy has taken for its domain questions concerning:
 - The nature of reality,
 - The analysis of reasoning, of justice, of beauty,
 - The study of moral values and principles,
 - The nature and sources of knowledge.

While neuroscience maps the brain, philosophy tries to map the mind.

Evolutionary Psychology and AI

- The intersection of EP and AI is a new frontier that promises to revolutionize our understanding of human behavior and cognition.
- By exploring the synergy between these two fields, scientists are unraveling the future of human-machine interaction and opening up new possibilities for the development of intelligent machines.
- Both fields are fundamentally concerned with understanding the mechanisms underlying intelligent behavior.

Src: [Exploring the Synergy between Evolutionary Psychology and Artificial Intelligence: Unraveling the Future of Human-Machine Interaction](#)



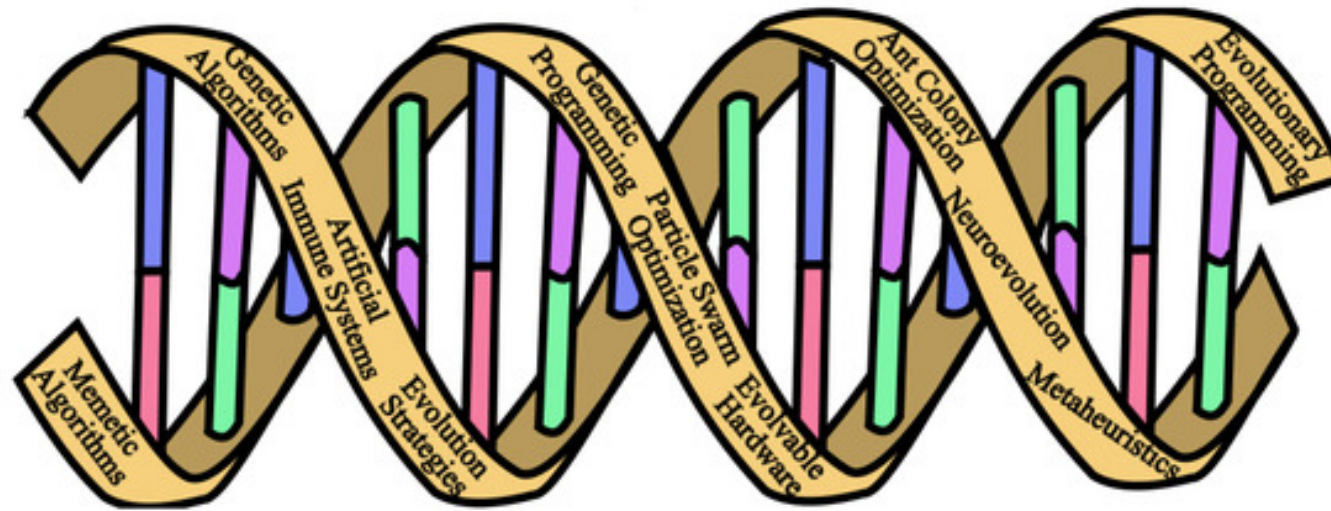
Evolutionary Psychology and AI

- Researchers can gain valuable insights into the nature of human cognition and develop more effective algorithms for solving complex problems.
- Possible area:
 - Development of AI systems that can understand and process human emotions.
 - Development of AI algorithms that are better able to mimic how humans learn and adapt to new situations and improve their performance over time
 - The intersection of evolutionary psychology and AI has the potential to shed light on some of the most pressing ethical and philosophical questions surrounding the development of intelligent machines.

Evolutionary Computing

- Evolution is a process that is equivalent to some mental operations, rather than simply as a collection of historical events that may have given rise to these operations.
- Recent years have seen the introduction of a new field of study that is known as evolutionary computing... three basic purposes
 - It solves real world problems, including problems having to do with financial forecasting, computer program design, and robot learning
 - Used to model and further understanding of natural evolutionary systems that exist within fields such as economics, immunology, and ecology
 - Serves as a metaphor for the operation of human thought processes.

Evolutionary Computing



There are now many forms of evolutionary computation that have developed over the years

These algorithms are designed so as to mimic certain behaviors as well as evolutionary traits of the human genome.

Evolutionary Computing

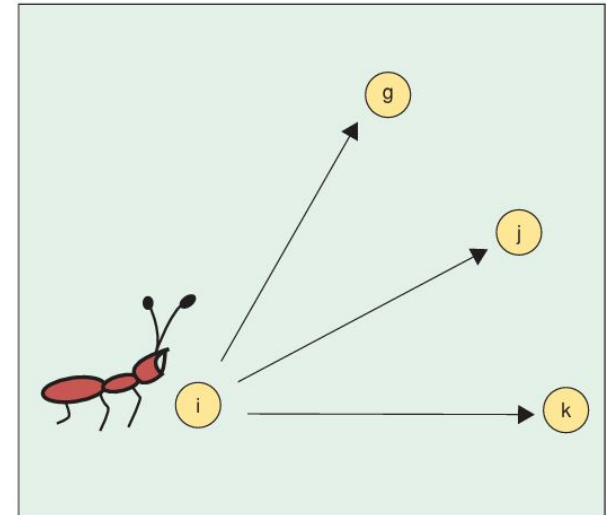
- All forms of EC share the same **general approach** to problem solving
 1. The generation of possible solutions or “candidate solutions”
 2. An evaluation of the fitness of the solutions
 3. A selection of the solutions that have the highest fitness values, based on some predetermined fitness threshold.
 4. A selection of the solutions that have the highest fitness values, based on some predetermined fitness threshold.
 5. Steps 2–4 are repeated, until the optimum solution is obtained.
- **EC cannot solve all problems well.**

Ant Colony Optimization

- **Swarm intelligence** is a relatively new approach to problem solving that takes inspiration from the social behaviors of insects and of other animals
- Ants have inspired a number of methods and techniques
 - Most studied and the most successful is the general purpose optimization technique known as ant colony optimization (ACO).
- Takes inspiration from the foraging behavior of some ant species.
- These ants **deposit pheromone** on the ground in order to mark some favorable path that should be followed by other members of the colony.

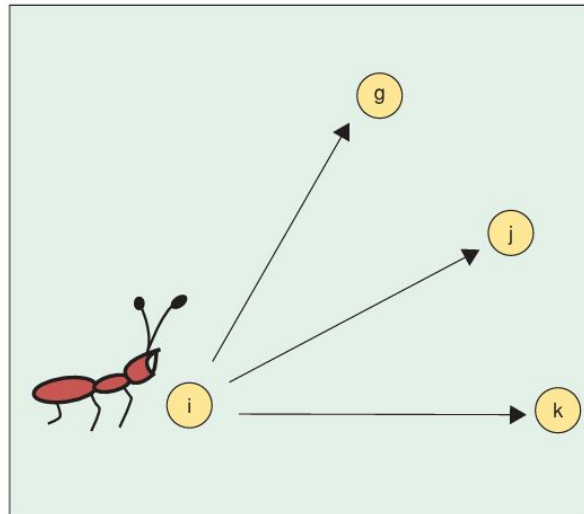
Ant Colony Optimization

- ACO is an iterative algorithm.
- At each iteration, a number of artificial ants are considered.
- Each of them builds a solution by walking from vertex to vertex on the graph with the constraint of not visiting any vertex that she has already visited in its walk.
- At each step of the solution construction, an ant selects the following vertex to be visited according to a stochastic mechanism that is biased by the pheromone:
 - When in vertex i , the following vertex is selected stochastically among the previously unvisited ones
- If j has not been previously visited, it can be selected with a probability that is proportional to the pheromone associated with edge (i, j)



Ant Colony Optimization

- ACO for the Traveling Salesman Problem
 - A set of cities is given and the distance between each of them is known.
 - **Goal:** find the shortest tour that allows each city to be visited once and only once.
 - The goal is to find a Hamiltonian tour of minimal length on a fully connected graph.
- In ant colony optimization, the problem is tackled by simulating a number of artificial ants moving on a graph (each vertex represents a city and each edge represents a connection between two cities.)



Ant Colony Optimization

Procedure AntColonyOptimization:

 Initialize necessary parameters and pheromone trials;

while not termination **do**:

 Generate ant population;

 Calculate fitness values associated with each ant;

 Find best solution through selection methods;

 Update pheromone trial;

end while

end procedure

Genetic Algorithm

- A genetic algorithm (or GA) is a search technique used in computing to find true or approximate solutions to optimization and search problems.
- Developed by John Holland, University of Michigan (1970's)
 - To understand the adaptive processes of natural systems
 - To design artificial systems software that retains the robustness of natural systems
- GAs are categorized as global search heuristics.
 - Global problem generally very difficult

Genetic Algorithm

- GAs are categorized as **global search heuristics**.
 - **Population-Based**: Unlike local search methods (like Hill Climbing) that move a single point, GAs maintain a population of hundreds of potential solutions.
 - **Parallel Exploration**: By having individuals scattered across the fitness landscape, GAs explore many regions of the search space simultaneously.
 - **Diversity**: Through Mutation, GAs can jump to entirely new areas of the map, preventing the algorithm from getting stuck in one small area.

Genetic Algorithm

- Use techniques inspired by evolutionary biology such as inheritance, mutation, selection, and crossover (also called recombination).
- Provide efficient, effective techniques for optimization and machine learning applications
- Widely-used today in business, scientific and engineering circles

Some GA Application Types

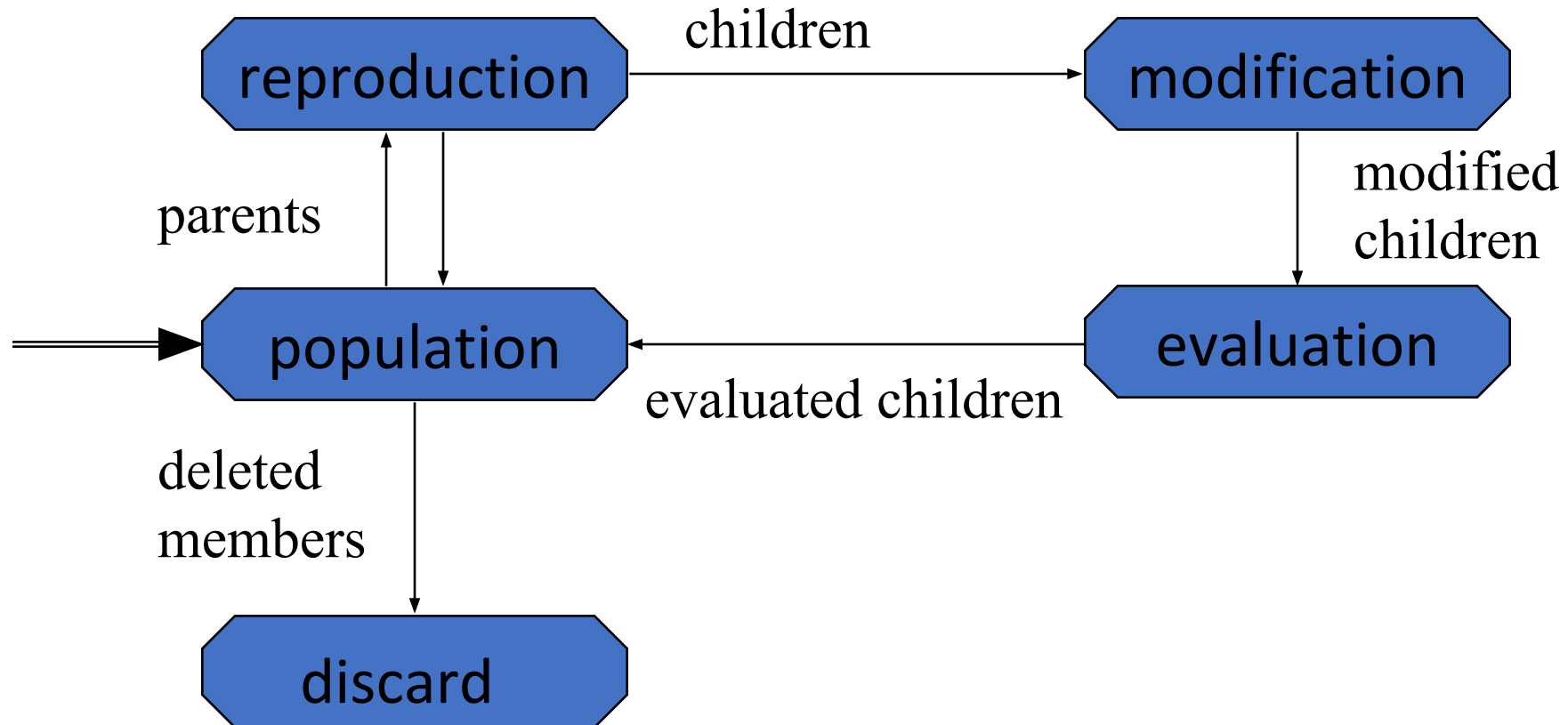
Expanded Domains and Application Types for Genetic Algorithms

Domain	Visualization	Application Types & Examples	
CONTROL		Real-time Systems Management (e.g., Gas Pipeline), Robotic Pole Balancing , Defense (e.g., Missile Evasion, Pursuit)	Autonomous Vehicle Fleet Management (e.g., routing, dispatching)
DESIGN		Complex Systems Design (e.g., Semiconductor Layout), Aerospace Engineering (e.g., Aircraft Design, component optimization), Ergonomics (e.g., Keyboard Configuration), Communication Networks	Material Science (e.g., creating novel material structures)
SCHEDULING		Manufacturing (e.g., Job-shop scheduling, production lines), Facility Scheduling (e.g., operating rooms, venues), Resource Allocation (e.g., personnel, inventory)	Public Transport (e.g., bus and train schedules)
ROBOTICS		Trajectory Planning and Optimization	Multi-robot Coordination and Swarm Algorithms
MACHINE LEARNING		Designing Neural Networks (e.g., architecture, hyperparameter optimization), Improving Classification Algorithms (e.g., feature selection), Classifier Systems	Transfer Learning (e.g., optimizing pre-trained models)
SIGNAL PROCESSING		Filter Design	Feature Extraction from Sensor Data
GAME PLAYING		Poker, Checkers, Prisoner's Dilemma	Strategy Optimization for Real-time Strategy (RTS) games
COMBINATORIAL OPTIMIZATION		Set Covering, Travelling Salesman, Routing, Bin Packing, Graph Colouring and Partitioning	Network Flow Maximization (e.g., internet traffic)

Genetic Algorithm: How it Works

- Start with a large “population” of randomly generated “attempted solutions” to a problem
- Repeatedly do the following:
 - Evaluate each of the attempted solutions
 - (probabilistically) keep a subset of the best solutions
 - Use these solutions to generate a new population
- Quit when you have a satisfactory solution (or you run out of time)

The GA Cycle of Reproduction



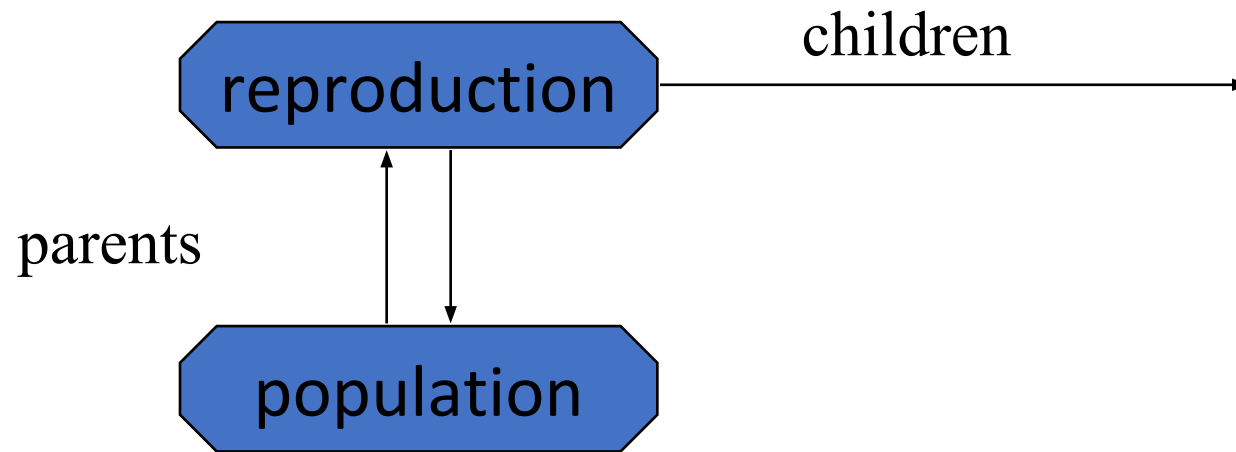
Population



Chromosomes could be:

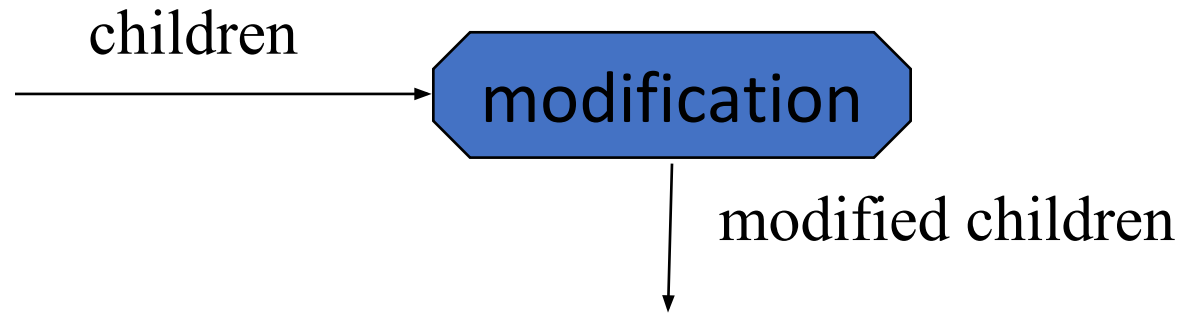
- Bit strings (0101 ... 1100)
- Real numbers (43.2 -33.1 ... 0.0 89.2)
- Permutations of element (E11 E3 E7 ... E1 E15)
- Lists of rules (R1 R2 R3 ... R22 R23)
- Program elements (genetic programming)
- ... any data structure ...

Reproduction



Parents are selected at random with selection chances biased in relation to chromosome evaluations.

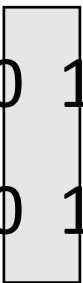
Chromosome Modification



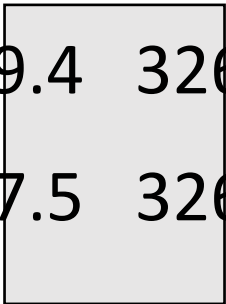
- Modifications are stochastically triggered
- **Operator** types are:
 - Mutation
 - Crossover (recombination)

Mutation: Local Modification

Before: (1 0 1 1 0 1 1 0)
After: (0 1 1 0 0 1 1 0)

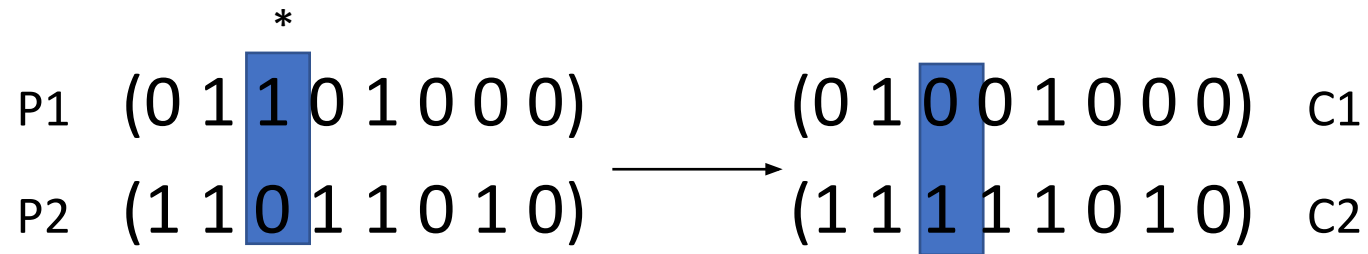


Before: (1.38 -69.4 326.44 0.1)
After: (1.38 -67.5 326.44 0.1)



- Causes movement in the search space
- Restores lost information to the population

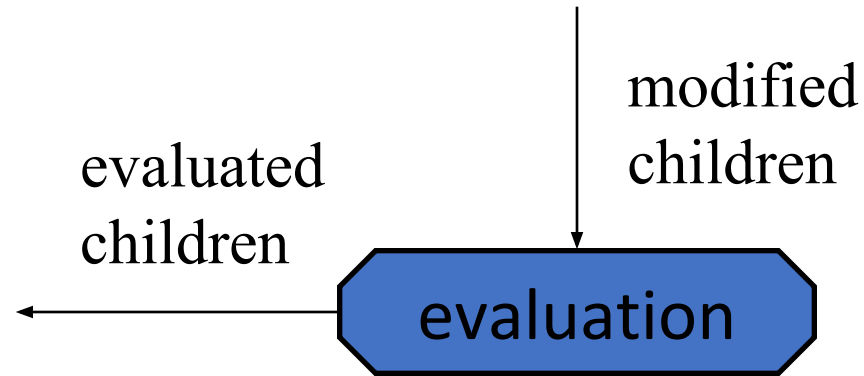
Crossover: Recombination



Crossover is a critical feature of genetic algorithms:

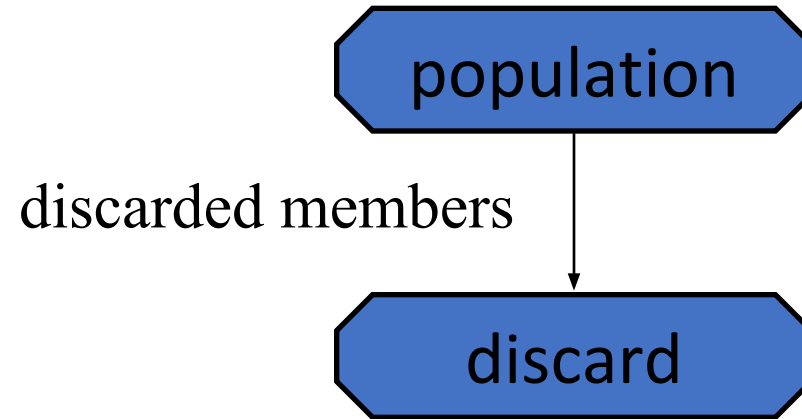
- It greatly accelerates search early in evolution of a population
- It leads to effective combination of schemata (subsolutions on different chromosomes)

Evaluation



- The evaluator decodes a chromosome and assigns it a fitness measure
- The evaluator is the only link between a classical GA and the problem it is solving

Deletion



- *Generational GA*: entire populations replaced with each iteration
- *Steady-state GA*: a few members replaced each generation

A Simple Example

The Traveling Salesman Problem:

Find a tour of a given set of cities so that

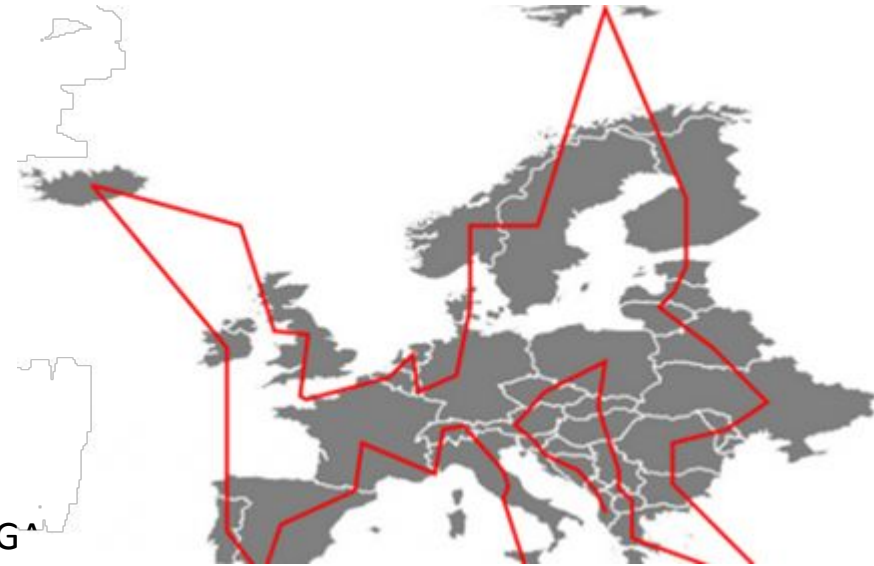
- each city is visited only once
- the total distance traveled is minimized

Representation is an ordered list of city numbers known as an *order-based G*

1) London 3) Dunedin 5) Beijing 7) Tokyo
2) Venice 4) Singapore 6) Phoenix 8) Victoria

CityList1 (3 5 7 2 1 6 4 8)

CityList2 (2 5 7 6 8 1 3 4)



Crossover

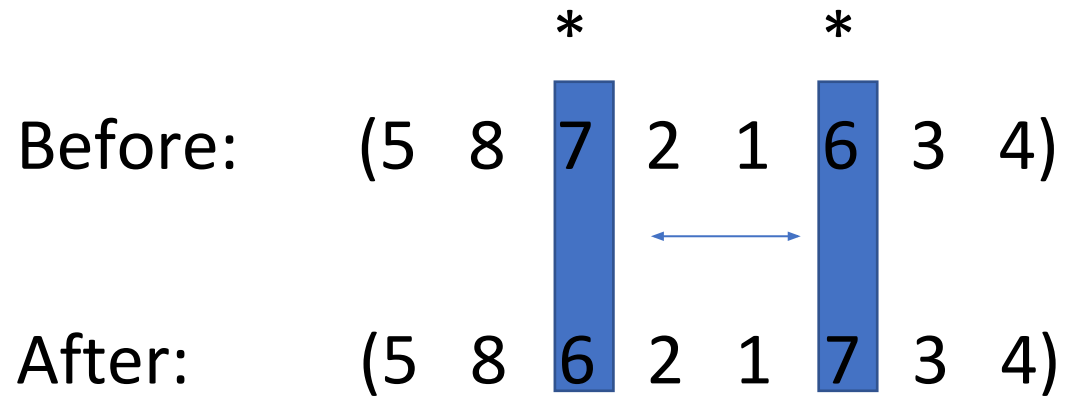
Crossover combines inversion and recombination:

			*		*		
Parent1	(3	5	7	2	1	6	4 8)
Parent2	(2	5	7	6	8	1	3 4)
Child	(5	8	7	2	1	6	3 4)

This operator is called the *Order1* crossover.

Mutation

Mutation involves reordering of the list:



Issues for GA Practitioners

- Choosing basic implementation issues:
 - Representation
 - Population size, mutation rate, ...
 - Selection, deletion policies
 - Crossover, mutation operators
- Termination Criteria
- Performance, scalability
- Solution is only as good as the evaluation function (often hardest part)